

Xinyu Li

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POSITION

University of Oxford

Sep 2026 – Present

Departmental Lecturer

Mathematical Institute, Mathematical and Computational Finance Group

University of Oxford

Sep 2025 – Aug 2026

Postdoctoral Research Associate

Mathematical Institute, Erlangen AI Hub

EDUCATION

University of California, Berkeley

09/2020 - 08/2025

Industrial Engineering and Operations Research (Ph.D.)

Thesis Title: An α -potential game framework for non-cooperative dynamic games: theory and algorithms

Advisor: Xin Guo

New York University Shanghai

09/2016-05/2020

Mathematics with *honors* (B.S.)

RESEARCH INTERESTS

Reinforcement Learning and Machine Learning Theory, Stochastic Control, Game Theory, Mean-Field Systems, Rough Path Theory, Mathematical Finance

INDUSTRY EXPERIENCE

Amazon, Santa Clara, CA

05/2023-08/2023 and 05/2022-12/2022

Middle Mile Planning, Research & Optimization Sciences

Mentor: Xin Guo, Philip Kaminsky

Efficient freight contract pricing using deep neural networks and transfer learning

Freight market rates forecast using machine learning models and signature transform

PUBLICATIONS

- X. Guo, X. Li, R. Xu, “Fast policy learning for linear quadratic control with entropy regularization”, *SIAM Journal on Control and Optimization*, 2026.
- X. Guo, X. Li, Y. Zhang, “An α -potential game framework for N -player dynamic games”, *SIAM Journal on Control and Optimization*, 2025.
- X. Guo, X. Li, C. Maheshwari, S. Sastry, M. Wu, “Markov α -potential games”, *IEEE Transactions on Automatic Control*, 2025.
- H. Gu, X. Guo, T.L. Jacobs, P. Kaminsky, X. Li, “Transportation marketplace rate forecast using signature transform”, *INFORMS Journal of Applied Analytics*, 2025.

Finalist of the 2024 *INFORMS Daniel H. Wagner Prize for Excellence in the Practice of Advanced Analytics and Operations Research*.

A short version was accepted by *30th SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024.

- Z. Zong, X. Li, P. Sanaei, “Effects of nutrient depletion on tissue growth in a tissue engineering scaffold pore”, *Physics of Fluids*, 2021.

WORKING PAPERS

- O. Bokanowski, J.F. Chassagneux, X. Li, C. Reisinger, “Non-asymptotic approximation and learning for path-dependent McKean–Vlasov control”, *preprint*, 2026
- X. Guo, G. He, X. Li, “Signature approach for contextual bandits with nonlinear and path-dependent rewards”, *arXiv preprint arXiv:2605.10313*, 2026.
- X. Guo, X. Li, Y. Zhang, “Distributed games with jumps: an α -potential game approach”, Minor version at *Mathematical Finance*, 2026.
- H. Gu, X. Guo, X. Li, “Adversarial training for gradient descent: analysis through its continuous-time approximation”, Revision at *Journal of Applied Probability*, 2025.

AWARDS AND SCHOLARSHIPS

- Bachelier Finance Society Travel Fund 2026
- Finalist, INFORMS Daniel H. Wagner Prize for Excellence in the Practice of Advanced Analytics and Operations Research 2024
- Berkeley Marshall-Oliver-Rosenberger Fellowship 2024
- UC Berkeley IEOR Departmental Fellowship 2020
- Zhang Xiaoqi & Cheung Kwok Ching Global Future Scholar (with ¥120k scholarship) 2018-2020

INVITED TALKS

- Department of Systems Engineering and Engineering Management Seminar, The Chinese University of Hong Kong, HK, China, 2026
- Applied Mathematics Seminar, The Hong Kong Polytechnic University, HK, China, 2026
- Oxford-ETH Workshop, ETH Zurich, Zurich, Switzerland, 2026
- International Conference on Computational Finance, Oxford, UK, 2026
- Stochastic Control and Reinforcement Learning, CIRM, Marseille, France, 2026
- World Congress of the Bachelier Finance Society, Bologna, Italy, 2026
- North British Probability Seminar, University of Edinburgh, Edinburgh, UK, 2026
- Oxford-PKU Joint Conference on Quantitative Finance and Data Science, Beijing, China, 2026
- Quantifying and Managing Uncertainty, Bielefeld, Germany, 2026
- Stochastic Finance Seminar, The University of Warwick, Coventry, UK, 2025
- PGMODAYS, EDF Lab Paris-Saclay, Palaiseau, France, 2025
- Bridging Stochastic Control and Reinforcement Learning: Theories and Applications, Newton Institute, Cambridge, UK, 2025
- 2023-2025 INFORMS Annual Meetings
- Oxford Mathematical and Computational Finance Seminar, Oxford University, Oxford, UK, 2025
- Machine Learning and Data Science Seminar, Oxford University, Oxford, UK, 2025
- Oxford-Princeton Workshop on Stochastic Analysis and Mathematical Finance, Oxford University, Oxford, UK, 2025

- SIAM Conference on Financial Mathematics and Engineering (FM25), Miami, FL, 2025
- Advances in Stochastic Control and Reinforcement Learning, Banff, Canada, 2025
- 2025 Western Conference on Mathematical Finance, USC, CA, 2025
- 2024 INFORMS Annual Meeting, Seattle, WA, 2024
- 2024 Joint Statistical Meetings, Portland, OR, 2024
- Banff Workshop on “New Trends and Challenges in Stochastic Differential Games”, Canada, 2024 (Online)
- Cyber Risk and Insurance France-Berkeley Conference, UC Berkeley, CA, 2024
- 58th Annual Conference on Information Sciences and Systems, Princeton University, NJ, 2024
- 4th ACM International Conference on AI in Finance, NY, 2023 (Online)
- 2023 INFORMS Annual Meeting, Phoenix, AZ, 2023
- International Congress on Industrial and Applied Mathematics, Waseda University, Tokyo, Japan, 2023 (Online)
- Berkeley-Columbia Workshop, Columbia University, NY, 2023
- Western Conference on Mathematical Finance, UC Berkeley, CA, 2023
- NYU Courant Summer Undergraduate Research Fund symposium, New York University, NY, 2019

ORGANIZER

- Co-organizer of the mini-symposium “Machine Learning for Decision Making in Finance” at INFORMS Conference on Financial Engineering and Fintech (Nov 2026)
- Co-organizer of the mini-symposium “Data-Driven Approaches to Stochastic Control and Games (I) and (II)” at International Conference on Computational Finance, Oxford, UK (Sep 2026)
- Co-organizer of the mini-symposium “Connections Between Mean Field Control and Machine Learning” at SIAM Annual Meeting, Ohio, USA (July 2026)
- Co-organizer of the workshop “Reinforcement Learning for Science: Discovery and Automation” at the Isaac Newton Institute Satellite Program, Cambridge, UK (March 2026)
- Organizer the Erlangen AI Hub bi-weekly seminars (Jan 2026 - Aug 2026)

REVIEWER

Journal of Machine Learning Research, SIAM Control and Optimization, SIAM Applied Mathematics, Mathematical Finance, Journal of Applied Probability, Applied Mathematics and Optimization, Applied Mathematical Finance, Finance & Stochastics, IEEE Transactions on Human-Machine Systems, NeurIPS 2025 GenAI in Finance Workshop

TEACHING EXPERIENCE

Graduate Student Instructor at UC Berkeley:

- IEOR 221 Introduction to Financial Engineering (Graduate), Spring 2025, Spring 2024
- IEOR 222 Financial Engineering System (Graduate), Spring 2023
- IEOR 242 Application in Data Analysis (Graduate), Spring 2022
- IEOR 142 Introduction to Machine Learning and Data Analysis, Fall 2022

Teaching Assistant at NYU Shanghai:

- Multivariable Calculus
- Introduction to Computer Programming